1.38 2D Matrices: Reflections

Worked Solutions

1 a)

Reflection in the x-axis.

1 b)

Reflection in the line $y = x$.

1 c)

Reflection in the y-axis.

1 d)

Reflection in the line $y = -x$.

2 a)

$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

2 b)

$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

2 c)

$\mathbf{A}^3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

2 d)

$\mathbf{A}^{2025} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

3)

The matrix represents a reflection in the y-axis. So the image will have the same area as the original shape. So the area of the shape's image is 4 units^2.

4 a)

$(3, 5) \mapsto (3, -5)$ is a reflection in the x-axis.

$\therefore \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

4 b)

$(-6, -2) \mapsto (6, -2)$ is a reflection in the y-axis.

$\therefore \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

4 c)

$(1, 9) \mapsto (-9, -1)$ is a reflection in the line $y = -x$.

$\therefore \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.

4 d)

$(-4, -7) \mapsto (-7, -4)$ is a reflection in the line $y = x$.

$\therefore \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

5)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 8 & 9 & 10 \\ 1 & 3 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & -3 \\ 8 & 9 & 10 \end{pmatrix}$$

So the coordinates of the vertices of T' are $(1, 8)$, $(3, 9)$ and $(-3, 10)$.

6)

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 6 & 11 & -1 \\ 5 & 5 & -3 & -3 \end{pmatrix} = \begin{pmatrix} -1 & 6 & 11 & -1 \\ -5 & -5 & 3 & 3 \end{pmatrix}$$

So the coordinates of the vertices of X' are $(-1, -5)$, $(6, -5)$, $(11, 3)$ and $(-1, 3)$.

7 a)

First reflect the point in the line $y = x$:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}$$

The vector from the origin to the closest point on the line is $\begin{pmatrix} -2.5 \\ 2.5 \end{pmatrix}$.

So the position vector of $(3, -5)$ when reflected in the line $y = x + 5$ is:

$$\begin{pmatrix} -5 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} -2.5 \\ 2.5 \end{pmatrix} = \begin{pmatrix} -10 \\ 8 \end{pmatrix}$$

$\therefore (-10, 8)$.

7 b)

First reflect the point in the line $y = -x$:

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$$

The vector from the origin to the closest point on the line is $\begin{pmatrix} 4 \\ 4 \end{pmatrix}$.

So the position vector of $(3, -5)$ when reflected in the line $y = 8 - x$ is:

$$\begin{pmatrix} 5 \\ -3 \end{pmatrix} + 2 \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 13 \\ 5 \end{pmatrix}$$

$\therefore (13, 5)$.

8 a)

$$\begin{pmatrix} \cos(2\times 30^\circ) & \sin(2\times 30^\circ) \\ \sin(2\times 30^\circ) & -\cos(2\times 30^\circ) \end{pmatrix} = \begin{pmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

8 b)

$$\begin{pmatrix} \cos(2\times 60^\circ) & \sin(2\times 60^\circ) \\ \sin(2\times 60^\circ) & -\cos(2\times 60^\circ) \end{pmatrix} = \begin{pmatrix} \cos 120^\circ & \sin 120^\circ \\ \sin 120^\circ & -\cos 120^\circ \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

8 c)

$$\begin{pmatrix} \cos(2\times 15^\circ) & \sin(2\times 15^\circ) \\ \sin(2\times 15^\circ) & -\cos(2\times 15^\circ) \end{pmatrix} = \begin{pmatrix} \cos 30^\circ & \sin 30^\circ \\ \sin 30^\circ & -\cos 30^\circ \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

8 d)

$$\begin{pmatrix} \cos(2\times 75^\circ) & \sin(2\times 75^\circ) \\ \sin(2\times 75^\circ) & -\cos(2\times 75^\circ) \end{pmatrix} = \begin{pmatrix} \cos 150^\circ & \sin 150^\circ \\ \sin 150^\circ & -\cos 150^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

9 a)

$\tan\theta = \sqrt{2} \Rightarrow \theta = \arctan\sqrt{2} = 54.7^\circ$ (3 SF).

$$\begin{pmatrix} \cos(2\arctan\sqrt{2}) & \sin(2\arctan\sqrt{2}) \\ \sin(2\arctan\sqrt{2}) & -\cos(2\arctan\sqrt{2}) \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} & \frac{2\sqrt{2}}{3} \\ \frac{2\sqrt{2}}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} -0.333 & 0.943 \\ 0.943 & 0.333 \end{pmatrix} \quad (3\text{ SF})$$

9 b)

$\tan\theta = 3 \Rightarrow \theta = \arctan 3 = 71.6^\circ$ (3 SF).

$$\begin{pmatrix} \cos(2\arctan 3) & \sin(2\arctan 3) \\ \sin(2\arctan 3) & -\cos(2\arctan 3) \end{pmatrix} = \begin{pmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{pmatrix} = \begin{pmatrix} -0.8 & 0.6 \\ 0.6 & 0.8 \end{pmatrix}$$

9 c)

$\tan\theta = \frac{1}{8} \Rightarrow \theta = \arctan\frac{1}{8} = 7.13^\circ$ (3 SF).

$$\begin{pmatrix} \cos(2\arctan\frac{1}{8}) & \sin(2\arctan\frac{1}{8}) \\ \sin(2\arctan\frac{1}{8}) & -\cos(2\arctan\frac{1}{8}) \end{pmatrix} = \begin{pmatrix} \frac{63}{65} & \frac{16}{65} \\ \frac{16}{65} & -\frac{63}{65} \end{pmatrix} = \begin{pmatrix} 0.969 & 0.246 \\ 0.246 & -0.969 \end{pmatrix} \quad (3\text{ SF})$$

9 d)

$\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \arctan\frac{1}{\sqrt{3}} = 30^\circ$.

$$\begin{pmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0.500 & 0.866 \\ 0.866 & -0.500 \end{pmatrix} \quad (3\text{ SF})$$

9 e)

$\tan\theta = -2 \Rightarrow \theta = 180° + \arctan(-2) = 117°$ (3 SF).

$$\begin{pmatrix} \cos(2(180° + \arctan(-2))) & \sin(2(180° + \arctan(-2))) \\ \sin(2(180° + \arctan(-2))) & -\cos(2(180° + \arctan(-2))) \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{pmatrix}$$

$$= \begin{pmatrix} -0.6 & -0.8 \\ -0.8 & 0.6 \end{pmatrix}.$$

9 f)

$\tan\theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = 180° + \arctan(-\frac{1}{\sqrt{2}}) = 145°$ (3 SF).

$$\begin{pmatrix} \cos(2(180° + \arctan(-\frac{1}{\sqrt{2}}))) & \sin(2(180° + \arctan(-\frac{1}{\sqrt{2}}))) \\ \sin(2(180° + \arctan(-\frac{1}{\sqrt{2}}))) & -\cos(2(180° + \arctan(-\frac{1}{\sqrt{2}}))) \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & -\frac{2\sqrt{2}}{3} \\ -\frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 0.333 & -0.943 \\ -0.943 & -0.333 \end{pmatrix} \quad (3 \text{ SF})$$

9 g)

$\tan\theta = -\frac{1}{5} \Rightarrow \theta = 180° + \arctan(-\frac{1}{5}) = 169°$ (3 SF).

$$\begin{pmatrix} \cos(2(180° + \arctan(-\frac{1}{5}))) & \sin(2(180° + \arctan(-\frac{1}{5}))) \\ \sin(2(180° + \arctan(-\frac{1}{5}))) & -\cos(2(180° + \arctan(-\frac{1}{5}))) \end{pmatrix} = \begin{pmatrix} \frac{12}{13} & -\frac{5}{13} \\ -\frac{5}{13} & -\frac{12}{13} \end{pmatrix} = \begin{pmatrix} 0.923 & -0.385 \\ -0.385 & -0.923 \end{pmatrix} \quad (3 \text{ SF})$$

9 h)

$\tan\theta = -\frac{1}{5\sqrt{3}} \Rightarrow \theta = 180° + \arctan(-\frac{1}{5\sqrt{3}}) = 173°$ (3 SF).

$$\begin{pmatrix} \cos(2(180° + \arctan(-\frac{1}{5\sqrt{3}}))) & \sin(2(180° + \arctan(-\frac{1}{5\sqrt{3}}))) \\ \sin(2(180° + \arctan(-\frac{1}{5\sqrt{3}}))) & -\cos(2(180° + \arctan(-\frac{1}{5\sqrt{3}}))) \end{pmatrix} = \begin{pmatrix} \frac{37}{38} & -\frac{5\sqrt{3}}{38} \\ -\frac{5\sqrt{3}}{38} & -\frac{37}{38} \end{pmatrix} = \begin{pmatrix} 0.974 & -0.228 \\ -0.228 & -0.974 \end{pmatrix} \quad (3 \text{ SF})$$

10 a)

$\cos 2\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{75°, 105°\}$

$\sin 2\theta = \frac{1}{2} \Rightarrow \theta = \{15°, 75°\}$

So $\theta = 75°$.

So the mirror line has equation $y = (\tan 75°)x \Rightarrow y = (2 + \sqrt{3})x \Rightarrow y = 3.73x$ (3 SF).

10 b)

$\cos 2\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{75°, 105°\}$

$\sin 2\theta = -\frac{1}{2} \Rightarrow \theta = \{105°, 165°\}$

So $\theta = 105°$.

So the mirror line has equation $y = (\tan 105°)x \Rightarrow y = (-2 - \sqrt{3})x \Rightarrow y = -3.73x$ (3 SF).

10 c)

$\cos 2\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{7.5°, 172.5°\}$

$\sin 2\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{7.5°, 82.5°\}$

So $\theta = 7.5°$.

So the mirror line has equation $y = (\tan 7.5°)x \Rightarrow y = 0.132x$ (3 SF).

10 d)

$\cos 2\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{37.5°, 142.5°\}$

$\sin 2\theta = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{37.5°, 52.5°\}$

So $\theta = 37.5°$.

So the mirror line has equation $y = (\tan 37.5°)x \Rightarrow y = 0.767x$ (3 SF).

11)

Midpoint of $(4, 6)$ and $(3 + 2\sqrt{3}, 2 - 3\sqrt{3})$ is $\left(\frac{7}{2} + \sqrt{3}, 4 - \frac{3}{2}\sqrt{3}\right)$.

Gradient from the origin to the midpoint is

$$\frac{4 - \frac{3}{2}\sqrt{3}}{\frac{7}{2} + \sqrt{3}} = 2 - \sqrt{3}$$

The angle the mirror line makes with the horizontal is $\theta = \arctan(2 - \sqrt{3}) = 15°$.

$$\therefore \begin{pmatrix} \cos(2 \times 15°) & \sin(2 \times 15°) \\ \sin(2 \times 15°) & -\cos(2 \times 15°) \end{pmatrix} = \begin{pmatrix} \cos 30° & \sin 30° \\ \sin 30° & -\cos 30° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

12)

Midpoint of $(-18, 20)$ and $(9 + 10\sqrt{3}, 10 - 9\sqrt{3})$ is $\left(-\frac{9}{2} + 5\sqrt{3}, 15 - \frac{9}{2}\sqrt{3}\right)$.

Gradient from the origin to the midpoint is

$$\frac{15 - \frac{9}{2}\sqrt{3}}{-\frac{9}{2} + 5\sqrt{3}} = \sqrt{3}$$

The angle the mirror line makes with the horizontal is $\theta = \arctan\sqrt{3} = 60°$.

$$\therefore \begin{pmatrix} \cos(2 \times 60°) & \sin(2 \times 60°) \\ \sin(2 \times 60°) & -\cos(2 \times 60°) \end{pmatrix} = \begin{pmatrix} \cos 120° & \sin 120° \\ \sin 120° & -\cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

13)

$\tan\theta = \frac{4}{1}$

$\Rightarrow \sin\theta = \frac{4}{\sqrt{17}}$ and $\cos\theta = \frac{1}{\sqrt{17}}$.

$\cos 2\theta = 2\cos^2\theta - 1 = 2\left(\frac{1}{\sqrt{17}}\right)^2 - 1 = -\frac{15}{17}$

$\sin 2\theta = 2\sin\theta\cos\theta = 2 \times \frac{4}{\sqrt{17}} \times \frac{1}{\sqrt{17}} = \frac{8}{17}$

$\therefore \mathbf{A} = \begin{pmatrix} -\frac{15}{17} & \frac{8}{17} \\ \frac{8}{17} & \frac{15}{17} \end{pmatrix}$.

14)

$\tan\theta = k$

$\Rightarrow \sin\theta = \frac{k}{\sqrt{1+k^2}}$ and $\cos\theta = \frac{1}{\sqrt{1+k^2}}$.

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = \left(\frac{1}{\sqrt{1+k^2}}\right)^2 - \left(\frac{k}{\sqrt{1+k^2}}\right)^2 = \frac{1}{1+k^2} - \frac{k^2}{1+k^2} = \frac{1-k^2}{1+k^2}$$

$$\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{k}{\sqrt{1+k^2}}\right)\left(\frac{1}{\sqrt{1+k^2}}\right) = \frac{2k}{1+k^2}$$

$$\therefore \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & -\frac{1-k^2}{1+k^2} \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}.$$